Problem Set 1 - Linear Algebra [1]

Due date: 08/25

This problem set reviews the key concepts covered so far. Please submit your assignment using LaTeX or R Markdown; this is meant to give you practice with these tools if you haven't used them before. Only one member of each group should email the submission to both instructors by the due date, with all other group members cc'ed.

1. Consider the following matrices:

$$\mathcal{A} = \begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix} \qquad \mathcal{B} = \begin{bmatrix} 7 & 2 \\ 6 & 3 \end{bmatrix}$$

   (a) Check $(\mathcal{A}\mathcal{B})^T = \mathcal{B}^T\mathcal{A}^T$.
   (b) Check $(\mathcal{A}\mathcal{B})^{-1} = \mathcal{B}^{-1}\mathcal{A}^{-1}$.
   (c) Check $(\mathcal{A}^T)^{-1} = (\mathcal{A}^{-1})^T$.

2. Let A and B be $n \times n$ matrices, where n is a positive integer $(n \geq 1)$. Prove whether the following is true or false:

$$\det(A + B) = \det(A) + \det(B)$$

3. For each function below, determine whether it is convex or concave using its Hessian matrix.
   (a) $f(x, y) = -x^2 - y^2 + xy + 2x - y$
   (b) $g(a, b, c) = 3a^2 + 2b^2 + c^2 - 2ab + 2bc - 6a - 4b - 2c$

4. Let X be a $n \times k$ matrix with $\text{rank}(\mathbf{X}) = k(n \geq k)$. An annihilator matrix $\mathcal{M}_{\mathbf{X}}$ is

$$\mathcal{M}_{\mathbf{X}} = \mathcal{I}_n - \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T.$$

   Show that $\mathcal{M}_{\mathbf{X}}$ is symmetric and idempotent. Clarify which property you are using in each step.

5. Find the eigenvalues and eigenvectors of the following matrix. Normalize the norm to one.

$$\mathcal{C} = \begin{bmatrix} .8 & .05 \\ .2 & .95 \end{bmatrix}$$

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6. Express the following terms
   a. $\sum_{i=1}^{n} \frac{u_i^2}{2\sigma^2}$
   b. $\sum_{i=1}^{n} \lambda_i u_i^2$

   into quadratic forms using the following notations:

$$\mathbf{u} := [u_1 \ \cdots \ u_n]', \quad \Sigma := \sigma^2 \mathcal{I}_n, \quad \text{and} \quad \Lambda := \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

7. Consider the following regression equation where $n > k$:

$$\begin{aligned} y_i &= x_{i1}\beta_1 + x_{i2}\beta_2 + \cdots + x_{ik-1}\beta_{k-1} + \beta_k + u_i && \forall\ i = 1, \cdots, n \\ &= {\mathbf{x}_i}^T \boldsymbol{\beta} + u_i && \forall\ i = 1, \cdots, n \end{aligned}$$

   We can rewrite this as a matrix equation

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$$

   where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \qquad \mathbf{X} = \begin{bmatrix} x_{11} & \cdots & x_{1k-1} & 1 \\ \vdots & & \vdots & \vdots \\ x_{n1} & \cdots & x_{nk-1} & 1 \end{bmatrix} \qquad \boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \qquad \mathbf{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}.$$

   (a) Check the following: $\mathbf{X}^T\mathbf{X} = \sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^T$ and $\mathbf{X}^T\mathbf{y} = \sum_{i=1}^{n} \mathbf{x}_i y_i$.

   (b) Assume that $\text{rank}(\mathbf{X}) = k$. We derived $\widehat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$ in class. Show that

$$(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \left(\frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^T \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i \mathbf{y}_i \right).$$

   (c) Show that

$$\sqrt{n}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \left(\frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^T \right)^{-1} \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i \mathbf{u}_i \right)$$

(d) Let $\hat{\mathbf{u}} := \mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}$. Show that $\sum_{i=1}^{n} \hat{u}_i^2 = \mathbf{u}^T \mathcal{M}_{\mathbf{X}} \mathbf{u}$ where $\mathcal{M}_{\mathbf{X}} = \mathcal{I}_n - \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$.

Note: You may start from $\hat{\mathbf{u}} = (\mathcal{M}_{\mathbf{X}})\mathbf{y} = \mathcal{M}_{\mathbf{X}}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u})$. You should be able to find that $\hat{\mathbf{u}} = (\mathcal{M}_{\mathbf{X}})\mathbf{u}$ and then do $\hat{\mathbf{u}}^T\hat{\mathbf{u}}$.

8. This is a problem using R.
Set up a function in R to determine the singularity of the matrix in each matrix equation below. The function should automatically derive a solution if it is non-singular, or should print "It does not have a unique solution" if a matrix is singular. (use `det`() function to derive the determinants of matrices. You can use the command `solve`() as well.)

(a)

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ -4 & 5 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ -9 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 4 & 2 \\ 2 & 5 & 1 \\ 3 & 6 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$$